

Tipping point analysis of atmospheric oxygen concentration

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We apply tipping point analysis to nine observational oxygen concentration records around the globe, analyse their dynamics and perform projections under possible future scenarios, leading to oxygen deficiency in the atmosphere. The analysis is based on statistical physics framework with stochastic modelling, where we represent the observed data as a composition of deterministic and stochastic components estimated from the observed data using Bayesian and wavelet techniques.

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We apply tipping point analysis to atmospheric oxygen concentration records. Experts warn that decrease of oxygen concentration from 21% to 19% would lead to significant health problems for humans and other animals, and we estimate the time scale of such a decline under various scenarios. We suggest that society should be careful with the introduction of new mass technologies that utilise double sink of atmospheric oxygen by both combustion and air-consuming synthesis.

I. INTRODUCTION

The increasing carbon dioxide concentration has been the major concern of the research community and policy-makers trying to address the challenges of global warming, as it was recently reported in the Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (Hartmann *et al.*, 2013). Less attention has been paid to atmospheric oxygen concentration, as it is considered much slower decaying and its proportional changes less dangerous for human population and biota. For instance, Keeling (1988) estimated that at contemporary fuel consumption atmospheric O_2 would be exhausted in 64 000 yr. Indeed, this time scale does not look threatening in comparison with rapid increase of carbon dioxide, but it is necessary to remember that human health problems appear much earlier than oxygen exhausts in full. Moreover, the past estimations of the decline trend may be overoptimistic, as the technological changes and population growth speed up consumption of fossil fuels and oxygen-consuming production of fertilisers and other materials.

There exist known effects of adaptation to high altitudes (Moore and Regensteiner, 1983; Frisanco, 2013), where people can survive at much lower concentrations of oxygen (heights up to 2500 m above sea level), with related changes in blood chemistry and oxygen transport system, especially during developmental period. These harsh conditions at high altitude may cause disease and poor health, but it is still possible to survive for humans: body systems adapt within several days or weeks due to physiological mechanisms, such as pulmonary ventilation, blood vessel

development, alveolar area, and oxygen transport increase (Frisanco, 2013).

Hypoxic conditions at birth lead to neurological and behavioural immaturity of newborns, as well as to smaller size, and infant mortality is twice as high at altitude. Adults may develop lung disease, high blood viscosity, and heart failure (Moore and Regensteiner, 1983). Still adaptation is successful in many cases (Frisanco, 2013).

It is important to note, however, that proportion of oxygen in air remains the same (about 20%) up to 110 km height, while air pressure drops: on the Mount Everest at 9 km oxygen (and air) pressure drop to almost 30% of the sea level pressure.

The situation is quite different when the proportion of oxygen in the air changes while pressure of the air remains the same. There are established regulations on oxygen deficiency published by the U.S. Centres for Disease Control and Prevention, where the dangerous level of oxygen quoted from the U.S. Occupational Safety and Health Administration (OSHA) and the U.S. National Institute for Occupational Safety and Health (NIOSH) are stated as 19.5% by volume (NIOSH Manual of Analytical Methods, 1994). If oxygen level drops to that value, employees are obliged to wear respiratory masks. If oxygen level drops to 16%–19%, it causes impaired thinking and coordination. Swaab (2013) states that a lack of oxygen prevents the brain *angula gyrus* from integrating the sensory information coming from the body, including the organs of balance, disrupting consciousness of the entire body. In some cases, it may even lead to delirium, which is the condition of loss of memory and perception in presence of major physiological functions.

The health and safety regulations for oxygen concentrations were developed primarily for an emergency in a confined space, in which the oxygen level drops suddenly. Obviously, in the open atmosphere of the planet, with the climatic changes at time scales of decades or longer, humans are likely to adapt gradually and maybe even flourish for some time if occupied only by meditation and basic life supplies, as it happens in high-altitude Tibet—probably due to reduced adrenalin level. Higher level of adrenalin requires more energy use by the organism, which is done via

adenosine triphosphate (ATP) in oxidative phosphorylation, and this leads to higher oxygen consumption in cells (Rich, 2003).

The problem of atmospheric oxygen deficiency should be addressed in advance, before drastic changes take place—compare this with the recent problem of the ozone hole. Too many environmental problems have been analysed by humanity in retrospect, and the purpose of this paper is to make an advance warning.

Yet another aspect discussed by Keeling (1988) is that the oxygen decay happens at time scale that is slower than decrease of estimated fossil fuel deposits. However, it is important to take into account rising technologies that may lead to mass production of new vehicles causing a double sink of atmospheric oxygen: not only due to combustion but also due to synthesis of intermediate materials. This may lead to a rapid increase of air consumption and much faster depletion of atmospheric oxygen.

The great oxygenation of the planet happened with appearance of photosynthesis about 2.5×10^9 years ago and later fluctuated until it stabilised at modern level, due to biochemical and geological processes (Canfield, 2014). Although oxygen depletion has been happening since the industrial revolution due to massive burning of fossil fuels, the meaningful and systematic observations started only a few decades ago, as it also required development of interferometric analysers. With development of observational instruments and building up of databases, ocean hypoxia became a topic of international research effort. It was reported that there are both natural and human-caused hypoxia of the world's coastal ocean (Rabalais *et al.*, 2010) as well as the interior ocean waters (Whitney *et al.*, 2007). The oxygen content of the ocean is defined by gas flux across air-sea interface and consumption due to microbial respiration (Falkowski *et al.*, 2011), which means that combined planetary oxygen budget is decreasing, and unless surface photosynthesis intensifies, we face global decline of oxygen.

In this paper, we study the currently available records of atmospheric oxygen using tipping point analysis, identify main components of the stochastic dynamical system, and project future development of the oxygen level.

Tipping points in climate system have recently become a widely publicised topic of high societal interest related to climate change; see, for example, Lenton *et al.* (2008). Applications of the tipping point analysis have been found so far in geophysics (Livina and Lenton, 2007; Lenton *et al.*, 2009; Livina *et al.*, 2010, 2011, 2012; Lenton *et al.*, 2012a, 2012b; Livina and Lenton 2013; Cimadoribus *et al.*, 2013; Livina *et al.* 2013), statistical physics (Vaz Martins *et al.*, 2010; Livina *et al.*, 2013), ecology (Dakos *et al.*, 2012), and structure health monitoring (Livina *et al.*, 2014).

Analysis of early warning signals of tipping points has generated a large number of important research papers, for instance, in ecology (see Scheffer *et al.* (2009), and references therein), whereas the early studies of noisy bifurcation precursors appeared already in the 1980s (Wiesenfeld, 1985). Although well-developed, these classical theories are difficult to apply in data-driven analysis of dynamical systems with unknown analytical description. Moreover,

societally relevant transitions (without structural change of the dynamical system) may be as important as bifurcations and require adequate analytical tools and techniques of analysis.

Whilst there is common understanding of tipping points as critical changes detectable in time series and attempts are made to classify tipping points (Thompson and Sieber, 2011; Ashwin *et al.*, 2012), there is a gap between the contemporary research literature on early warning signals that are being analysed with basic autocorrelation techniques and analytical bifurcation theory in dynamical systems of the second half of the 20th century (Arnold, 1992; Milnor, 1969). Tipping point analysis fills this gap and provides flexible instruments (analytical tools as well as software) for such analysis. The approach combines general analytical description of the dynamical system with data-driven derivation of the underlying system potential, which makes it applicable to a broad range of real-world dynamical systems.

One of the main reasons for that is that many models employed in the theory of bifurcations are purely deterministic (although chaotic), which makes them of restricted applicability for studying various real-world complex systems, where there is lack of knowledge of analytical description of the system. For example, climate system modelling, which started as physically based and deterministic, with years of development involved higher and higher number of variables and parameters and at certain level of complexity it was understood that its modelling performance can be matched by simpler and computationally lighter stochastic models (Hasselmann, 1976; Palmer and Williams, 2008). In fact, this is correct not only for climatic models but also for models of many other complex systems.

A stochastic model involving deterministic and stochastic components (static or dynamic) is a powerful yet simple tool for modelling arbitrary time series of real-world dynamical systems. These systems are characterised by complexity and high number of variables producing a single system trajectory (for example, a temperature records influenced by hundreds of other climate variables like humidity, wind speed, etc). In most cases, it is impossible to reconstruct a deterministic model from such recorded datasets, and the analysis of the system has to be data-driven. This makes analysis of tipping points even more complicated, and novel techniques are needed to address this challenge. The main advantage of such stochastic approach is that the multitude of the complex factors influencing the dynamics of the system are separated into two major components, with smaller-scale fluctuations under the umbrella of the stochastic term. This allows one to monitor the major variability of the system state, combined with analysis of the shorter-term fluctuations dynamics.

Given one-dimensional trajectory of a dynamical system (recorded time series), the system dynamics may be modelled by the following, very general, stochastic equation with state variable z and time t :

$$\dot{z} = D(z, t) + S(z, t), \quad (1)$$

where D and S are deterministic and stochastic components, correspondently. Component $D(z, t)$ may be stationary or

include a trend, depending on the studied system. The probability distribution of such a system, however complex it may be, in majority of cases can be approximated by a polynomial of even order (so-called potential system). The stochastic component, in the simplest case, may be Gaussian white noise, although in real system it is often more complex than that possessing correlations (red noise), anti-correlations (blue noise), multifractal, and other nonlinear properties.

Tipping points can be defined and classified in terms of the underlying system potential whose derivative, if exists, defines the deterministic term in Eq. (1), i.e., $D(z, t) = -U'(z, t)$. If the potential structure (number of potential wells) changes, the tipping point is a genuine bifurcation. If the potential structure remains the same, while the trajectory of the system samples various states, such a tipping point is transitional. An example of such transition may be the record of global temperature, which has the same structure of fluctuations with drifting (under forcing or noise-induced). It is important to identify and distinguish types of tipping points (bifurcations vs transitions) for the purpose of their further forecast. This is a key question in order to define the horizon of the forecast and general predictability of the tipping point (whether it is abrupt or gradual).

The methodology has general applicability for studying trajectories of dynamical systems of arbitrary origin and serves to anticipate, detect and forecast tipping points. It is currently based on the techniques of degenerate fingerprinting and potential analysis, which are described in the Methodology section.

II. DATA

There is a long-term observational project at the Scripps Institute that monitors the oxygen anomaly in several stations around the globe (Keeling, Scripps O_2 Program, 2014). Oxygen measurements are reported as changes in the $\frac{O_2}{N_2}$ ratio of sampled air relative to a reference (air pumped in the mid-1980s stored in the Scripps laboratory). The unit is “per meg,” which means that a decline of 1 per meg is equal to 0.0001% decline of oxygen concentration. Because natural sources and sinks of N_2 are much smaller than those of O_2 , the changes in $\frac{O_2}{N_2}$ ratio mostly reflect changes in O_2 (Keeling, Scripps O_2 Program, 2014).

The Scripps O_2 datasets have variable timescale, with average time interval of two weeks. Their length varies from 391 (Cold Bay) to 688 (Mauna Loa) datapoints. For the purposes of our wavelet analysis, we interpolated all series to obtain number of datapoints 4096 (power of 2, which is important for applying synchrosqueezing), equidistant in time.

The observed data of the Scripps O_2 Program (Keeling, 2014) is shown in Figure 1.

III. METHODOLOGY

A. Stochastic model with heteroscedasticity

(Chen *et al.*, 2014) proposed a modelling approach for studying time series with periodicities and trends, which is based on the synchrosqueezing wavelet transform (Daubechies *et al.*, 2011) implemented in a publicly available Matlab Synchrosqueezing Toolbox (Brevdo, 2014; Thakur *et al.*, 2013).

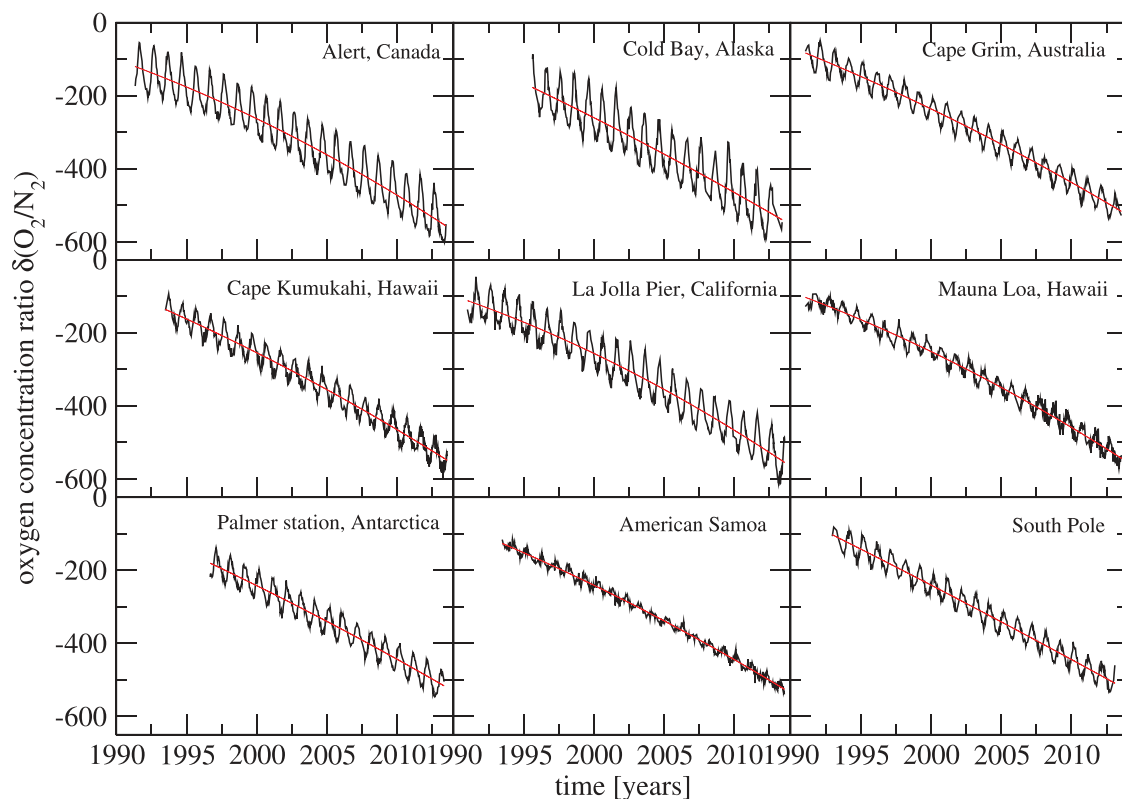


FIG. 1. Observed oxygen concentration data from nine stations around the globe. Unit is the change of $\frac{O_2}{N_2}$ ratio in per meg, 0.0001% of decline of oxygen concentration. The least-square quadratic fit is shown by red line.

Following [Chen et al. \(2014\)](#), we approximate the observed oxygen data by stochastic model

$$Z(t) = T(t) + A(t) \cos(2\pi\varphi(t)) + \tau(t)\Phi(t),$$

where trend $T(t)$ is a real-valued function, $\Phi(t)$ is a stationary generalised random process, and $\tau(t)$ is a positive real-valued smooth function used to model the heteroscedasticity of the error term. As an example, $\Phi(t)$ can be continuous autoregressive moving average random process of order (p, q) . This model was assumed when [Livina et al. \(2013\)](#) modelled declining Arctic sea ice as a composition of parabolic trend, seasonal periodicity and fluctuations. Similarly, [De Livera et al. \(2011\)](#) used trigonometric Box-Cox transform with ARMA errors and seasonal components.

The synchrosqueezing wavelet transform allows one to model variable seasonality in the data as a sum of intrinsic mode functions (IMFs)

$$s(t) = \sum_{k=1}^K A_k(t) \cos(\varphi_k(t)). \quad (2)$$

Synchrosqueezing wavelet transform is based on discrete wavelet transform with chosen mother wavelet in composition with standard circular Discrete Fourier Transform, see [Daubechies et al. \(2011\)](#) and [Thakur et al. \(2013\)](#). We deploy the synchrosqueezing wavelet transform (2) for derivation of the time-varying seasonal trend in the oxygen data.

B. Trend estimation

In Figure 1, one can see that a few of the nine datasets have visibly parabolic decline. The simplest way to remove such a trend is to subtract the least-square parabolic fit from the data. However, Bayesian inference ([Sivia and Skilling, 2006](#)) provides an attractive approach to model comparison, which can be utilised in this case to define whether the trend is quadratic or linear. It allows one to focus on the quantities of interest by integrating out other parameters, and the incorporation of previous information into the evaluation of the evidence in favour of a model. Importantly, Bayesian inference eliminates the danger of overfitting, by automatically penalising model complexity. This is a real advantage when trying to ascertain whether oxygen has been declining linearly or quadratically over the last decades. Unless the additional parameter of the quadratic model significantly increases, the probability of the data given our prior information—the evidence—the linear model is preferred. The analytical models considered are linear $y = Ax + B \sin(\omega x) + C \cos(\omega x) + D$ and quadratic $y = Ax^2 + Bx + C \sin(\omega x) + D \cos(\omega x) + E$.

The ratio of evidences between two models (the Bayes factor) expresses the degree to which the data supports one model over the other. However, resulting from integration over the entire parameter space, its evaluation may represent a great numerical challenge. To overcome this problem, [Skilling \(2004\)](#) proposed nested sampling to compute evidence, an efficient Monte Carlo technique that samples uniformly from the prior, subject to a hard constraint that the

likelihood of the new sample point is greater than that of the previously discarded point.

Nested sampling ([Skilling, 2004](#)) has found applications in various areas ([Aitken and Akman, 2013](#); [Dybowski et al., 2013](#); [Pullen and Morris, 2014](#); [Murray et al., 2006](#); [Jasa and Xiang, 2012](#); [Mukherjee et al., 2006](#); [Feroz et al., 2009](#)), and here we employ it to compare the evidence in favour of a quadratic versus linear model, for each data set of oxygen decline. We use Jeffreys' scale ([Jeffreys, 1961](#)) to interpret the Bayes' factors; the results support the hypothesis that oxygen is declining parabolically in the following oxygen datasets: Alert, Cape Grim, La Jolla, Mauna Loa, American Samoa.

Furthermore, we test several types of polynomial models (orders from 1 to 4, which include parabolic $Ax^2 + Bx + C$ and linear $Ax + B$), logarithmic $A + B \log x$ and exponential $-\exp Ax + B$, using information criteria, Bayesian BIC, Akaike AIC and Akaike corrected AICc, for model selection ([Hurvich and Tsai, 1989](#); [Burnham and Anderson, 2004](#)). Information criteria allow one to select the most plausible model when they reach minimal value in the model range considered. We apply this analysis to annually averaged oxygen data, and Fig. 2 shows the results, where both linear and exponential models are discarded in favour of higher-polynomial-degree models, for which we take quadratic model as the most feasible selection.

Information criterion AICc was shown to perform better than AIC and BIC for small-size datasets ([Yang and Forbes, 2011](#)). This can be illustrated in the analysis of annually averaged oxygen data, when the original oxygen time series reduce to 20+ datapoints.

We repeated the analysis of information criteria for high-resolution interpolated oxygen data (not shown), and Table I provides summary of model selection for three type of data for nine considered datasets. Quadratic decline of oxygen is supported by majority of dataset.

To obtain average over the projections of nine oxygen records, we use equal weights with assumption of quadratic decline trend (Fig. 8).

C. Tipping point analysis

The tipping point analysis consists of the following three stages:

- anticipating (pre-tipping)
- detecting (tipping)
- forecasting (post-tipping)

Anticipating tipping points (pre-tipping or early warning signal) is based on the effect of slowing down of the dynamics of the system. When a system state becomes unstable and starts a transition to some other state, the response to small perturbations becomes slower. This signal of "critical slowing down" is detectable as increasing autocorrelations (ACF) in the time series ([Held and Kleinen, 2004](#)).

Alternatively, the short-range scaling exponent of Detrended Fluctuation Analysis, DFA, may be monitored. The lag-1 autocorrelation value is calculated in sliding windows of fixed (conventionally, half of the series length, or

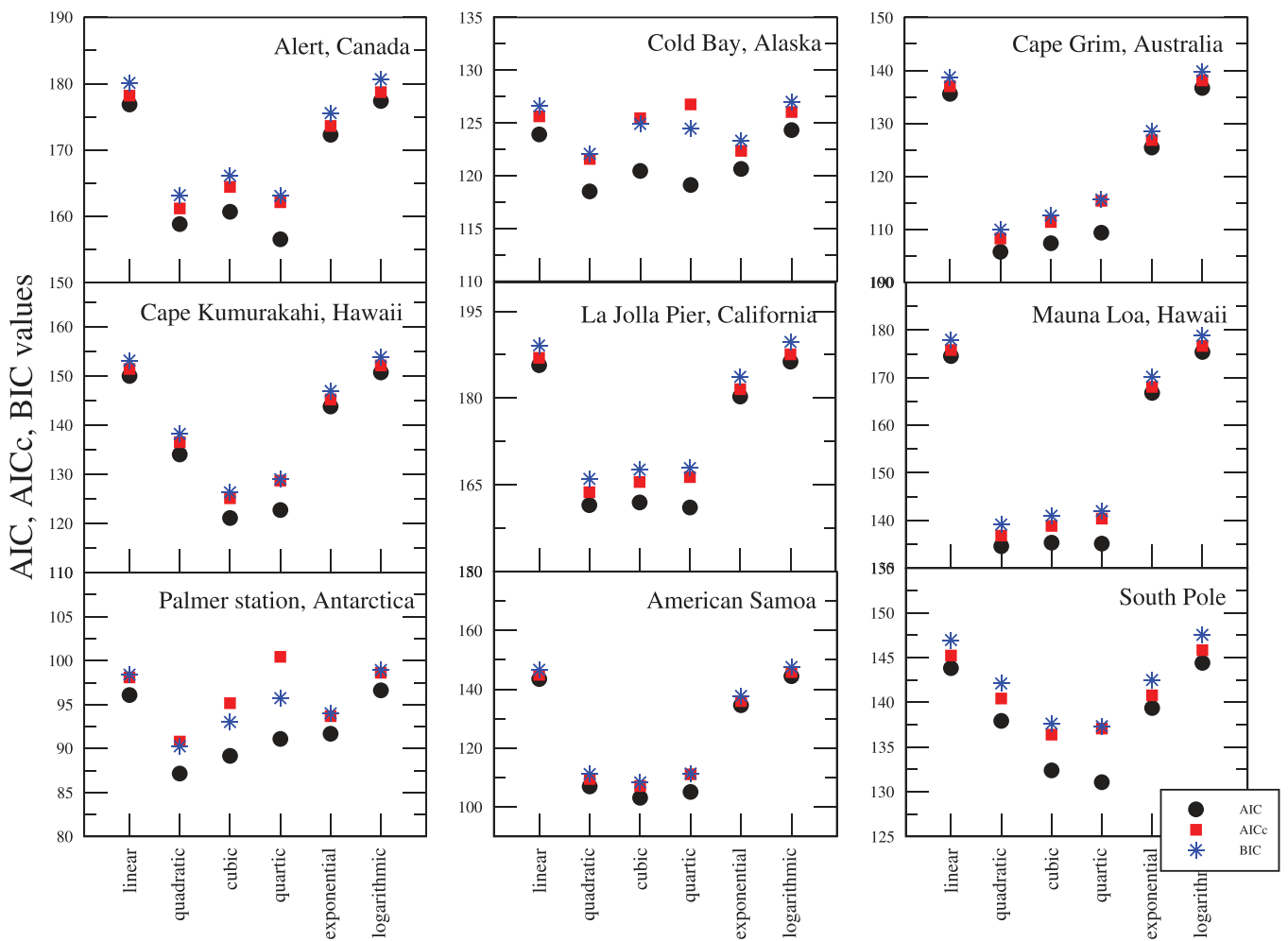


FIG. 2. Information criteria for annually averaged data. The annually averaged oxygen data select higher-degree polynomial models, which means that linear decline of oxygen is very unlikely. Note that in all nine series neither the linear nor the exponential model are ever selected; all of the datasets select a polynomial model of order between 2 and 4, and we choose quadratic model (parabolic decline of oxygen) as a conservative approximation.

TABLE I. Summary of selected models for three types of oxygen data: initial data with unequal time steps; interpolated equidistant data, and annually averaged data. While raw data may suggest linear decay in case of three stations, the rest of data, including interpolated and annually averaged, confirms quadratic model selection, based on information criteria BIC, AIC, and AICc.

	Raw data		Interpolated data		Annual average data	
	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic
Alert		✓		✓		✓
Cold Bay	✓			✓		✓
Cape Grim		✓		✓		✓
Cape Kumurakahi		✓		✓		✓
La Jolla Pier		✓		✓		✓
Mauna Loa		✓		✓		✓
Palmer Station	✓			✓		✓
American Samoa		✓		✓		✓
South Pole	✓			✓		✓

variable length for uncertainty estimation) along a time series. Combined together, these dynamically derived values form a curve of an early warning indicator whose pattern describes the behaviour of a time series. If the curve of the indicator remains flat and stable, this means that the time

series does not experience any critical change (whether bifurcational or transitional). If the indicator rises to a critical value 1 (the monotonic rise is assessed using Kendall rank correlation), it provides a warning of critical behaviour.

Lag-1 autocorrelation is estimated by fitting an autoregressive model of order 1 (linear AR(1)-process) of the form:

$$z_{t+1} = c \cdot z_t + \sigma \eta_t$$

where η_t is a Gaussian white noise process of unit variance, and the “ACF-indicator” (AR1 coefficient) is as follows:

$$c = e^{-\kappa \Delta t}$$

where κ is the decay rate of perturbations, and $c \rightarrow 1$ as $\kappa \rightarrow 0$, while a tipping point is being approached.

Note that for the purposes of degenerated fingerprinting (Held and Kleinen, 2004), it is possible to follow the conventional general procedure of fitting an auto-regressive model to the dataset under consideration, but the lag-1 autocorrelation value can also be derived directly from the time series, because we approximate the series with the chosen autoregressive model. For AR(1) process, $X_t = \phi_1 X_{t-1} + \varepsilon_t$, autocovariance $\gamma_k = \text{cov}(X_t, X_{t-k}) = \text{cov}(\phi_1 X_{t-1} + \varepsilon_t, X_{t-k}) = \phi_1 \gamma_{k-1}$. This means that $\gamma_k = \phi_1^{|k|} \gamma_0$, $k \in \mathbb{Z}$. Since the empirical

autocorrelation function is always normalised (i.e., $\gamma_0 = 1$), it means that lag-1 autocorrelation γ_1 coincides with the AR(1) parameter ϕ_1 and can be obtained directly from the ACF of the time series.

Alternatively, DFA method utilises a built-in detrending of prescribed polynomial order, which allows one to distinguish transitions and bifurcations in the early-warning signals. This can be done using several early warning indicators, with and without detrending data in sliding windows (Livina *et al.*, 2012).

The paper (Livina and Lenton, 2007) provided the first application of the DFA-based early warning indicator to the paleotemperature record with detected transition using both ACF and DFA indicators. The result was the implementation of detrending in the assessment of short-term auto-correlations, using the built-in DFA detrending as well as linear detrending in sliding windows combined with lag-1 autocorrelation estimation.

Ditlevsen and Johnsen (2010) argue that between 25 and 45 kyr BP, the Greenland paleotemperature did not experience a system bifurcation but rather oscillated between stable system states. It is an interesting confirmation of the result of Livina *et al.* (2010), where by using a different technique, the potential contour plot, it was shown that in the mentioned time interval the system potential has stable double-well structure. The same can be concluded using DFA indicator (but not the ACF-indicator, which does rise to critical value of 1, together with variance indicator, and thus provides false alarm of a tipping point that does not happen, see Figure 3).

Detecting and forecasting of a tipping point is performed using dynamical potential analysis. The potential analysis detects a transition or a bifurcation in a series at the time when it happens, which is illustrated in innovative contour plot mapping the potential dynamics of the system (Livina *et al.*, 2010; 2011). The dynamics of the tipping point is then forecast using extrapolation of the dynamically derived Chebyshev coefficients of the approximation of the probability density (Livina *et al.*, 2013).

For the purposes of potential analysis, the dynamics of the system is approximated by a simple stochastic model with a polynomial potential U

$$\dot{z} = -U'(z) + \sigma\eta$$

where $\dot{z}$ is the time derivative of the system variable $z(t)$, η is Gaussian white noise of unit variance, and σ is the noise level. In the case of a double-well potential, U is a polynomial of 4th order

$$U(z) = a_4 z^4 + a_3 z^3 + a_2 z^2 + a_1 z.$$

According to the Fokker-Planck equation for the dynamic evolution of the probability density function $p(z, t)$

$$\partial_t p(z, t) = \partial_z [U'(z)p(z, t)] + \frac{1}{2} \sigma^2 \partial_z^2 p(z, t),$$

its stationary solution is given by

$$p(z, t) \sim \exp[-2U(z)/\sigma^2].$$

The potential can be reconstructed from time series data of the system as

$$U(z) = -\frac{\sigma^2}{2} \log p_d(z),$$

which means that the empirical probability density p_d (kernel distribution) has the number of modes corresponding to the number of wells of the potential.

It is important to be able to identify the structural change of the potential in the recorded trajectory of dynamical system. Such changes are often subtle and invisible to the eye, yet leading to dramatic evolution of the system. The potential contour plot visualises bifurcations according to the number of detected system states. It identifies bifurcations as the change in the number of states along all time scales (y-axis shows the size of the sliding window with subset of data where the number of states is assessed). If no such pattern is observed, there is no bifurcation in the series. If the potential plot is “patchy,” indicating constantly varying potential of the system, it can be classified as non-potential, i.e., no global deterministic quasi-stationary potential exists to describe the dynamics of the system.

This simple approximation works with remarkable accuracy for data subsets of length as short as 400–500 data points, demonstrating above 90% rate of successful detection, as was shown in an experiment with double-well-potential artificial data: for data subsets of length above 1000 points, it correctly detects the structure of the potential with rate above 98% (Livina *et al.*, 2011). In the real-world dynamical systems, the recorded trajectories are often very short, and high reliability of the method in the case of poor statistics is a clear advantage.

The technique of **potential forecasting** is based on dynamical propagation of the probability density of time series. We employ polynomial coefficients of the orthogonal approximation of the empirical probability distribution and extrapolate them in order to forecast the future probability distribution of data. After reconstruction of the system kernel distribution, a time series is generated using rejection sampling technique, and then the obtained dataset is sorted according to the initial data in order to reconstruct the temporal correlations in the series.

For deterministic models with a linear perturbation of the system potential, it is possible to analyse the proximity to a bifurcation (Surovyatkina, 2005), but the problem of the real-world systems is that the potential may be completely unknown, and the perturbation may not be linear. In fact, all terms of the polynomial potential may vary simultaneously. Such complex dynamics means that a possible way of analysing the system is to derive the sequence of the potential coefficients and attempt to identify the leading parameter that perturbs the system potential. The pilot study in this direction was performed in the paper (Livina *et al.*, 2013) where the coefficients of the approximation of kernel distribution were used for the system forecast.

This technique is useful for detailed forecast of the fluctuations dynamics at short-term scale. However, in the current paper we focus on the long-term projection of oxygen

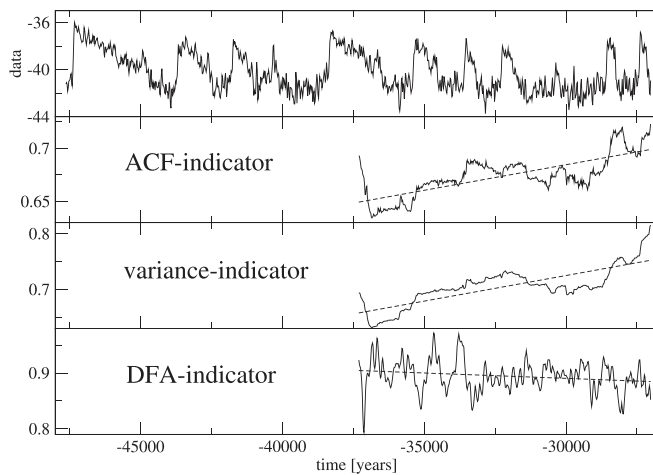


FIG. 3. Greenland Ice Core Project (GRIP) paleotemperature proxy in time interval 45–25 kyr BP, where there is no system bifurcation, which is confirmed by DFA-indicator—whereas ACF-indicator and variance give false alarm of tipping (dashed lines show the slopes of linear regressions, with indication of early warning signal in ACF and variance, but not in DFA-indicator). The result is confirmed by different techniques (Livina *et al.*, 2010; Ditlevsen and Johnsen, 2010).

decline (on the scale of millennia), which is driven, as we show below, by the trend rather than by fluctuations, and therefore sub-annual dynamics does not need to be forecast.

IV. RESULTS AND DISCUSSION

We remove parabolic trend from the data (least-square polynomial fit of 2nd order) and apply synchrosqueezing wavelet transform with Morlet mother wavelet, 32 voices (parameter that reflects the number of scales of decomposition) and threshold parameter $\gamma = 1.4901 \times 10^{-8}$ (Brevdo, 2014). The number of IMFs used is 3. By summing up the obtained empirical decomposition components, we reconstruct variable seasonal trend in the data. Note that (Chen *et al.*, 2014) utilised continuous wavelet transform for estimation of trends in their methodology, assuming trend variability along the studied time series, but in the case of oxygen the trend can be well approximated by a single function, which we obtain as least-square parabolic fit. Yet another reason why this trend is used is that we need to perform extrapolation of the oxygen series, for which we use analytical description of the trend rather than a series. In Fig. 4, we show four years of detrended data (Alert, Canada) and its synchrosqueezing seasonal function calculated with three IMFs. Fluctuations can then be derived as difference of these two series. For comparison, we also performed conventional pre-processings of the oxygen series, first by detrending and deseasonalising (i.e., subtracting average seasonal cycle) and then by deseasonalising and detrending.

We apply detrending and synchrosqueezing, alongside conventional detrending and deseasonalising for comparison, and show the derived fluctuations in Fig. 5. In further analysis, we use fluctuations derived using synchrosqueezing.

Early warning ACF-indicators are shown in Fig. 6, and most of them do not indicate critical behaviour in fluctuations, being stable horizontal or declining, providing no early warning for transitional behaviour, except very short-term change

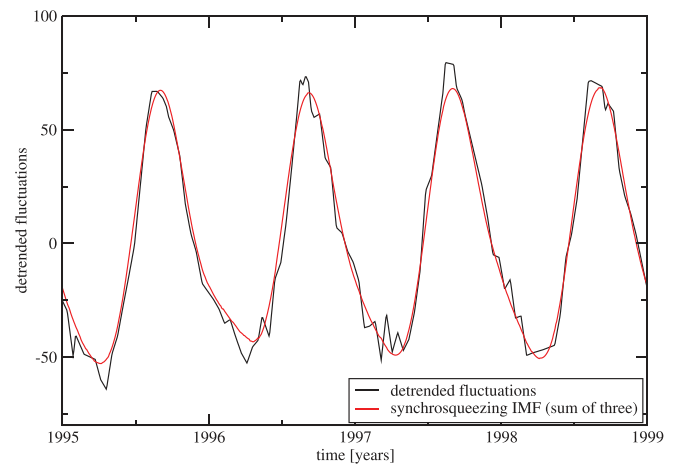


FIG. 4. Detrended ALT data and sum of its synchrosqueezing IMFs (first three functions) used for deseasonalising of fluctuations.

at the end of series of Alert and Palmer. This could be an interesting effect to explore when longer series are available for analysis in a few years' time. The indicators have error-bars that are derived from multiple curves of indicators calculated for window sizes varying from $1/4$ to $3/4$ of series length.

Potential contour plots are shown in Fig. 7, and they are generally monotonous one-well-potential, which means that fluctuations themselves (after detrending and synchrosqueezing) do not have structural changes. Therefore, the dynamics of the oxygen series can be represented as simple composition of trend, seasonal variation, and red-noise fluctuations (fluctuation exponent above 1, not shown here).

It is interesting to compare potential contour plots with early warning indicators and the fluctuation series themselves: once the short-term changes are detected in the potential contour plot, they are easier tracked in the fluctuations and in short-term dynamics of indicators.

By combining these results, we can conclude that the detrended fluctuations of the oxygen series do not have a current tipping point, and therefore the transitional behaviour is trend-defined.

We use the derived analytical trend (parabolic) to project the long-term decline of the oxygen concentration. Given the exponential growth of population and consumption, the trend is unlikely to be linear and as slow as (Keeling, 1988) estimated to be dozens of thousand years. In Fig. 8, we show the projection of the trend (without taking into account fluctuations and seasonality) which may lead to health-threatening problems for humans and other animals in one thousand years or sooner. The error-bars are shown as a shadow of the projection, based on trend variations of the nine considered records. We propose parabolic decline as conservative estimate of the oxygen dynamics: although in the short observational data the estimated trend is not exponential, the current population growth and related consumption of fuels, air oxygen and emissions may well be exponentially growing and soon leading to faster than parabolic decline.

Although the original data in Fig. 1 shows oxygen declining with linear or parabolic trend, such short observations may be misleading, as Fig. 9 with world population hindcast demonstrates.

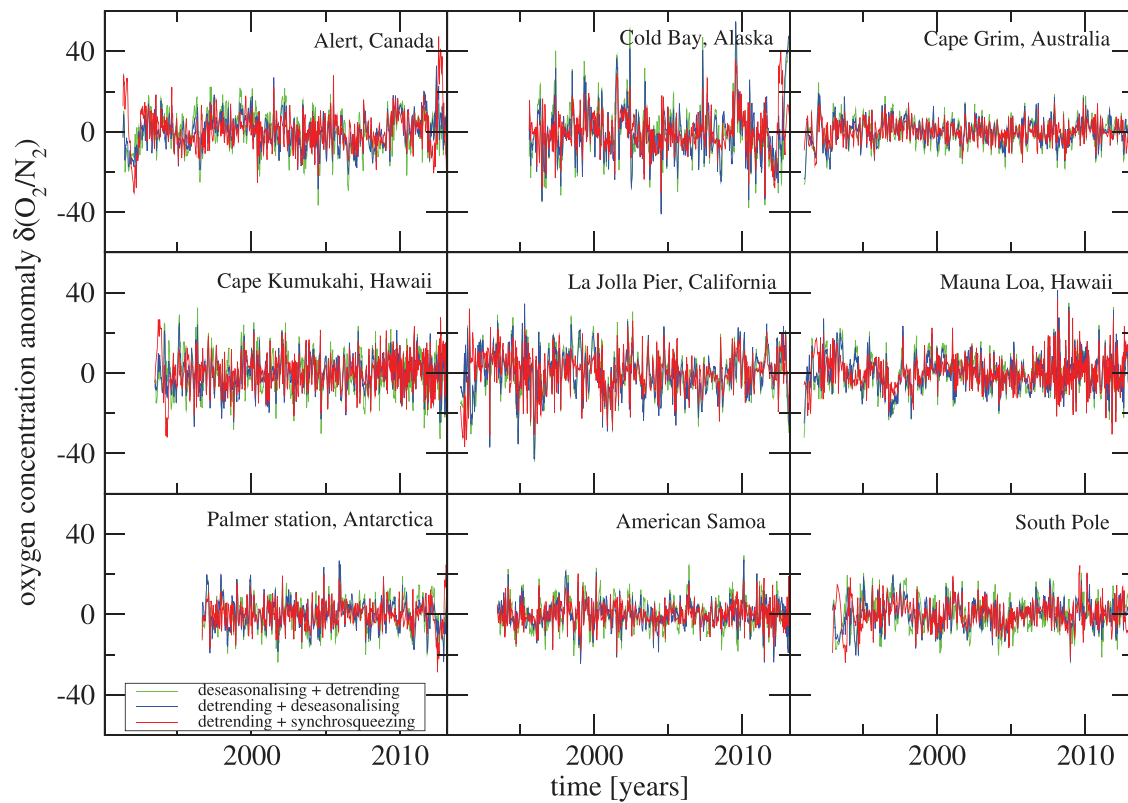


FIG. 5. Fluctuations obtained after three different pre-processing techniques: (1) deseasonalising + detrending; (2) detrending + deseasonalising; (3) detrending + synchrosqueezing. Units of obtained data are ratio of oxygen and nitrogen, per meg, as in Fig. 1.

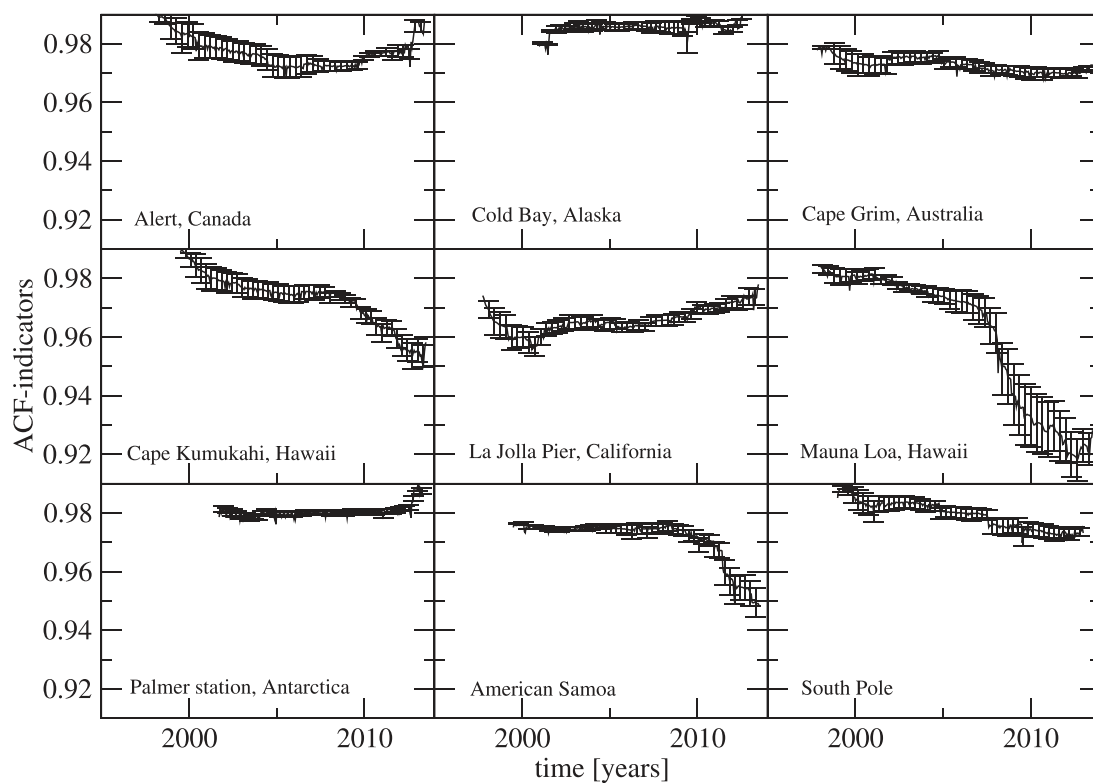


FIG. 6. ACF-indicators of fluctuation time series. There is no signal of critical behaviour, except may be the Palmer Station in Antarctica, at the very end of the series. High values of indicators are due to the interpolation for synchrosqueezing.

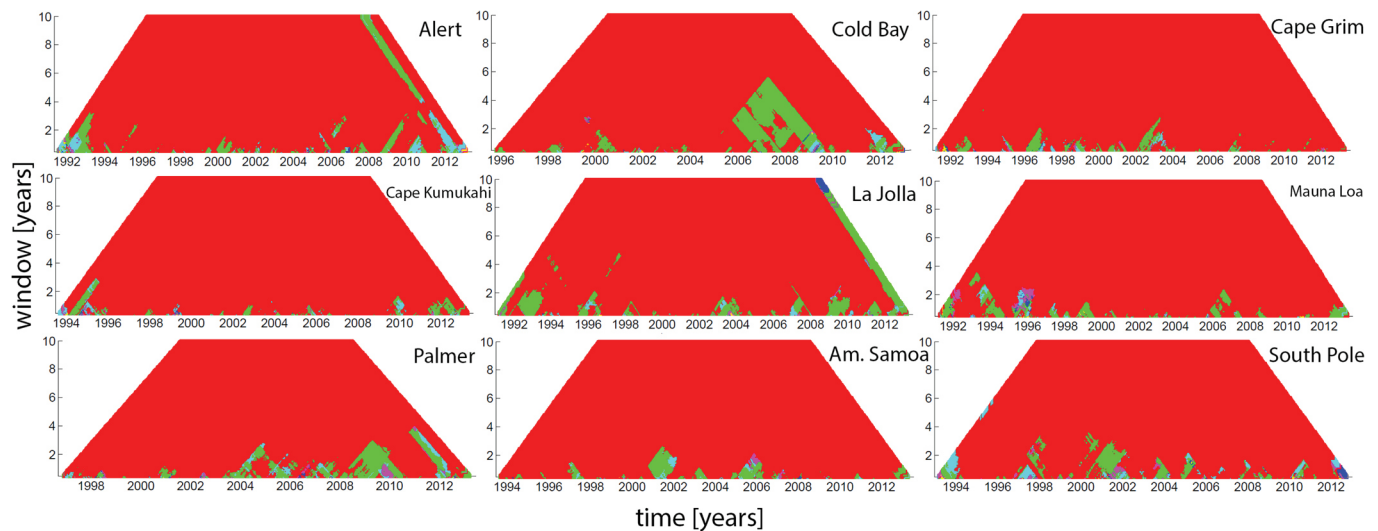


FIG. 7. Potential contour plots for oxygen ratio fluctuations (anomaly), red colour denotes one-well-potential system and green—double-well-potential. The series are generally one-well-potential, with some short-term variations at small time scales.

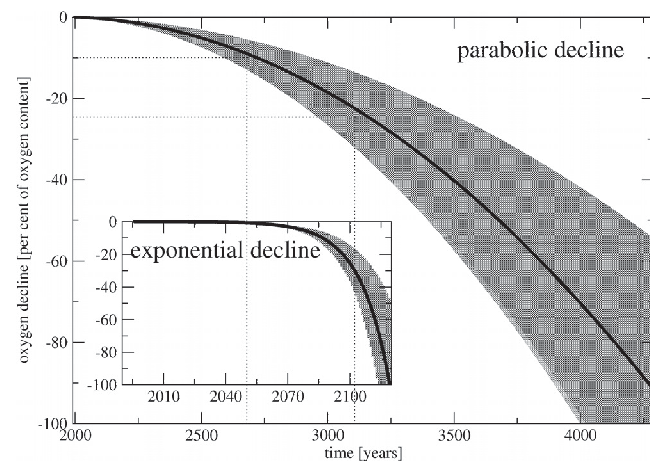
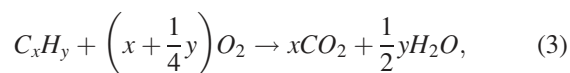


FIG. 8. Projection of the oxygen decline using extrapolation of derived trends in the oxygen data, averaged over parabolic least-square fits. The scale in the y-axis denotes decline of oxygen in per cent of its current volume, which means that decline of 2% of atmospheric content, when health problems start, corresponds to about 10% decline in this plot; 5% decline of atmospheric content corresponds to 25% decline in this plot (these levels are shown by dotted lines, with corresponding time scales). The inset with exponential decline is shown for illustrative purposes; we consider realistic the parabolic decline in the main frame.

Keeling (1988) provides a general formula for burning fossil fuels that deplete the atmospheric oxygen



which means that every mole of fossil fuel requires $x + 0.25y$ moles of oxygen, depending on the fuel components. The common natural gas used for energy generation is methane CH_4 (requires 2 mol of oxygen per mole of fuel), while car petrol is C_8H_{18} (requires 12.5 mol of oxygen per mole of fuel). Due to the molar mass ratio (32 for oxygen and 44 for carbon dioxide), fuel combustion according to Eq. (3) would take 0.4 kg of oxygen to produce 0.352 kg of carbon dioxide, i.e., the amount of the emitted carbon dioxide is a proxy of

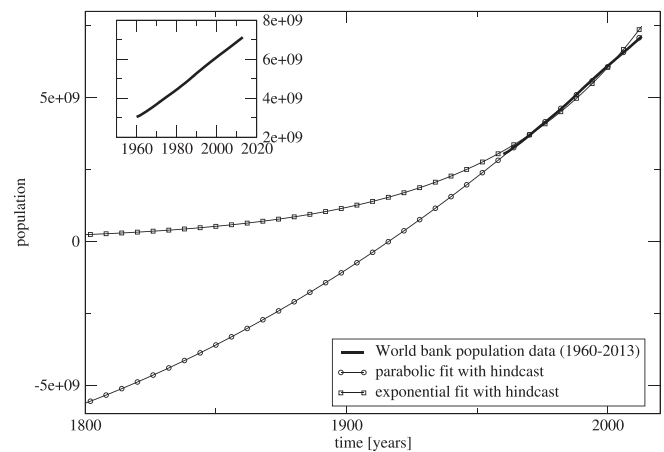


FIG. 9. World population data (thick curve in the main plot and in the inset) and backward hindcasts of population using parabolic and exponential least-square fits. Although the trend in the inset curve (1960–2013) looks linear, even parabolic fit is not realistic for approximating its dynamics at the scale of a few centuries.

the amount of consumed oxygen in this chemical process alone.

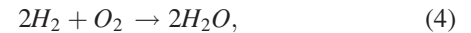
Keeling considered highly detailed derivations of oxygen consumption factors, but what he could not know back in 1988 is that new mass technologies would emerge, as well as the current population growth and related level of resources consumption were not taken into account then. Unfortunately, the recent funding reduction of the USA research programs affected the Scripps O_2 program (Tollefson, 2013), although the earlier linear estimates of the oxygen decline may be well too optimistic and require constant monitoring. Given the above estimation of almost equal use of oxygen in producing carbon dioxide in fossil fuel burning, it is obvious that the established exponential rise of carbon dioxide should lead to accelerating (i.e., not linear) decline of oxygen.

But it is not only direct fossil fuel burning that may reduce the oxygen level. There are various indirect factors of

great uncertainty that may cause accelerated oxygen depletion. For example, if we assume that world population stabilises and stops growing exponentially, and transport use may stop growing, oxygen may still continue to decline due to increase of soil fertilisation and production of synthetic materials. In Fig. 10, one can see that in the last 10 years, the use of fertilisers increased by 35% per hectare of arable land. This one factor alone may become critical in terms of oxygen depletion within a few decades, if the dynamics remain the same. (Ciais *et al.*, 2007) reported previously unconsidered anthropogenic nitrogen-related mechanisms with impact on atmospheric oxygen trends which drive the global nitrogen cycle to a more oxidised state through nitrification and nitrate fertilizer industrial production. Similarly, Randerson *et al.* (2006) hypothesized that recent changes in biomes have caused carbon within the terrestrial biosphere to become increasingly more oxidised over a period of decades, which means speeding up of atmospheric oxygen decline. Sirignano *et al.* (2010) studied records of two stations in the Netherlands and Ireland and detected annual trend acceleration of oxygen decline over the period of observation. These results support our estimation of the oxygen dynamics as declining faster than linear.

The atmosphere has a mass of about 5.13×10^{18} kg (Trenberth and Smith, 2005) of which oxygen represents 23.15%, about 1.19×10^{18} kg. The statistics of the World Bank (World Bank Data Portal, 2014) reports global carbon emissions from transport as rising from 2.5 to 6×10^{12} kg since 1970, and increase of the number of cars per 1000 people around the globe from 100 to 130 since 2000, while population at the same time (since 2000) increased from 6 to 7.1×10^9 . This means that exponential growth of population will lead to exponential growth not only of carbon emissions but also of air oxygen consumption, unless the industrial-scale processes start monitoring this. With the current trend of replacement of direct fossil fuel combustion by new technologies like fuel cells, carbon emissions may be restricted, but not necessarily the use of air oxygen.

A hydrogen fuel cell has a net reaction



but in addition to this use of oxygen, hydrogen production requires oxygen directly or indirectly, either in high temperature-pressure technologies (using fossil fuels) or in fertilisation for production of biofuels. This means that in the current industrial conditions hydrogen fuel technologies maintain double sink of atmospheric oxygen, because they utilise oxygen when hydrogen is synthesised, and combust oxygen when utilise fuel energy.

Cement and fertilisers production, petrochemical products of various types (oils, waxes, greases, explosives, rubbers, alcohols, asphalts), fuel cell operation and manufacturing, energy and heat generation—all require air oxygen for their industrial cycles. It is not clear how these processes can be curtailed at global scale. For example, Haber-Bosch process ($N_2 + 3H_2 \rightarrow 2NH_3$) for production of nitrogen fertilisers uses methane from natural gas as source of hydrogen, at a temperature of 300–500 °C and pressure of 150–250 bars, which in addition requires energy (which in its part is often generated by gas turbines).

In order to illustrate the potential impact of the recent mass technologies (those developed on industrial scale during the 20th century) on oxygen depletion, we show in Fig. 10 several global variables (World Bank Data Portal, 2014).

This figure shows observed variables for the last 40–50 years, and the data are not long enough to estimate whether the trends are linear, parabolic, or exponential (as the population growth).

Although there were several publications forecasting stabilisation of the population growth, a recent paper (Gerland *et al.*, 2014) states that by the end of the 21st century the world population may reach 12×10^9 or higher. To address the growing goods consumption, one of the main world manufacturers, China, according to (Jones *et al.*, 2014), has demand for ethylene increased from 7.2 MM m.t. in 2005 to 16.4 MM m.t. in 2013, with estimation of

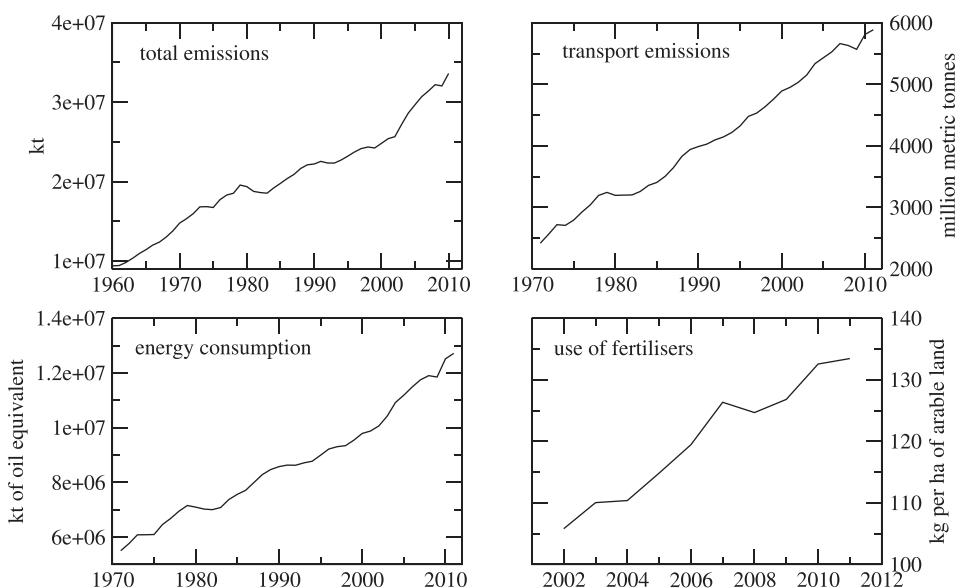


FIG. 10. Four variables of the world statistics (World Bank Data Portal 2014) showing clear increase in resources consumption, which all cause oxygen depletion. Note increase of use of soil fertilisers by 30% in the past 10 years; doubled energy consumption in the past 40 years; tripled transport emissions in the past 40 years.

28.3 MM m.t. in 2017. When process volume doubles every 5 yr, this gives indication for at least parabolic increase, if not exponential.

Therefore, as conservative approximation, taking into account increasing consumerism, parabolic trends of oxygen depletion can be adopted at least for medium term predictions, which justifies the estimations of oxygen decline scale shown in Fig. 8.

In the context of the tipping point analysis, atmospheric oxygen concentration undergoes transitional (not bifurcational) post-tipping. This means that the system potential does not change in shape but rather drifts with time. Because the instrumental observations of oxygen concentration started very recently, we have no data to detect the past tipping point itself, which was likely in the mid-19th century. In our opinion, this tipping was purely anthropogenic, caused by the industrial revolution and accelerated by globalisation of technological processes and population growth. In the recent study (Edgeworth *et al.*, 2015), the authors provide a time scale of human impact on the Earth, starting from 13 800 BP, and argue that the industrial revolution that started in the late 18th century was greatly accelerated in 1945–1950, with A-bomb-derived radionuclides serving as the marker of the onset of the Anthropocene acceleration and rapid rise in multiple indicators of anthropogenically induced change in global systems (Steffen *et al.*, 2011). It is important to note that although the industrial revolution started in the 18th century, only by 1850 did the CO₂ concentration reach the upper limit of natural Holocene variability, which can be seen in the widely published record of this variable that may serve as a combustion-related anti-correlated proxy of oxygen dynamics. Moreover, although the oxygen measurement data earlier than 1980s is not available, the decline of atmospheric oxygen, if estimated back into past thousands of years with the current dynamics, would imply much higher level of oxygen over Holocene, which is not confirmed by paleorecords. Therefore, it is likely that the tipping point in the atmospheric oxygen took place in the past 100–150 years, and we observe its post-tipping dynamics.

V. CONCLUSION

The importance of oxygen for humans and animals is obvious. It is possible that with slow rate of increasing oxygen deficiency human population will gradually adapt to new conditions, as it happened in the high altitude areas around the globe. However, if some new mass technologies with oxygen sink would be adopted by the world consumers (in addition to the industrial combustion processes), this may drastically increase oxygen deficiency, eventually leading to critical misbalance of the atmosphere. The scale of the change will depend on the population growth, consumerism speed and international technological policies. If left uncontrolled, the declining oxygen concentration could lead to serious threats to animal life, including the human race.

Currently, in innovative cars like the dual-fuel BMW Hydrogen 7 (BMW 2014), 170 l of liquid hydrogen provide driving range of 200 km, while petrol tank of the same car with 73 l of gasoline gives range of 480 km. Taking into

account molar masses and net equations (3) and (4), this technology uses about 2.7 times less volume of oxygen for combustion in hydrogen mode than in petrol mode. Nowadays, hydrogen fuel is often synthesised by non-green generation, and this innovative car in the present conditions does maintain a double sink of oxygen. However, if in future hydrogen generation is powered by renewable sources of energy, such technologies might be a feasible solution for reducing transport-related oxygen depletion.

It is important to remember that even if all fossil fuels are fully exhausted, and the “dirty” internal combustion engines are replaced by hydrogen fuel transport (or similar), the planet will continue losing oxygen at the rate of the growing number of transport vehicles. A future-technology scheme that would combine green generation and storage of fuel with increased efficiency of engines may address this technological challenge.

We suggest that all industrial processes introducing new technologies with air consumption should be assessed regarding oxygen depletion before they are deployed at full scale.

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